

**"AN ANALYSIS ON PURCHASE OF RAW MATERIALS,
PRODUCTION, SALES OF KAMDHENU NXT TMT BARS AND
ITS FINANCIAL OPERATIONS - A CASE OF
HINDUPUR STEEL AND ALLOYS (P) LTD."**

**A Summer Training Report submitted to
ISME, BANGALORE
in partial fulfilment of the requirements
for the award of the
POST GRADUATE DIPLOMA IN MANAGEMENT**

**Submitted by
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**Under the guidance of
Dr. Chithra S, Professor (Faculty Guide)
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**International School of Management Excellence
Bangalore**

CERTIFICATE

This is to certify that the Summer Training Report entitled: “**An Analysis on purchase of raw materials, production, sales of Kamdhenu NXT TMT Bars and its Financial Operations – A Case Study of Hindupur Steel and Alloys (P) Ltd.**” submitted in partial fulfilment of the requirements for the award of Post Graduate Diploma in Management is a record of original training undergone by Mr. K. Mohammed Abdulla Khan (Reg. No. 2511421) during his course of study at International School of Management Excellence, Bangalore. The report has been prepared under my supervision and has not formed the basis for the award of any Degree/Fellowship or similar title to any candidate of any University.

Place: Bangalore

Date: 10-06-2026

Signature of Faculty Guide

Dr. Chithra S.

Professor.

Countersigned

Principal
ISME, Bangalore



HINDUPUR STEEL & ALLOYS PVT. LTD.

(Authorised Manufacturer of Kamdhenu Hvt under Licence user agreement)
(An ISO Certified Company)

Regd. Off : 1341 , A Block, RMV Extension, Sahakarnagar, Bangalore - 560092.
Works:- Plot No. 29, Industrial Park, Gollapuram Village, Hindupur Mandal, Sri Sathya Sai Dist (A.P.) - 515 211

CIN : U27100KA2009PTC049915

JAS-ANZ



CM/JL-0004708468

CERTIFICATE

This is to certify that Mr. K. Mohammed Abdulla Khan, a student of Post Graduate Diploma in Management, 2025-2027, International School of Management Excellence, Bangalore bearing Enrolment No 2511421, has undertaken the Summer Internship Training at Hindupur steels and Alloys (P) Ltd. During 01-Apr-2026 to 31-May-2026 under my supervision & guidance. He has conducted a study & completed the Project on 5-day purchase, production & sales review.



Seal of Organization

Date: 04-06-2026

Designation: General Manager

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Mr. Pradip Kumbhakar

DECLARATION

I, **K. Mohammed Abdulla Khan** , hereby declare that the Summer Training Report, entitled **“An Analysis on purchase of raw materials, production, sales of Kamdhenu NXT TMT Bars and its Financial Operations – A Case Study of Hindupur Steel and Alloys (P) Ltd”** , submitted to the International School of Management Excellence, Bangalore, in partial fulfillment of the requirements for the award of the Post Graduate Diploma in Management is a record of original training undergone by me during the period **April- May 2019** Under the supervision and guidance of Your **Guide name with qualification and designation, International** School of Management Excellence, Bangalore **and** it has not formed the basis for the award of any Degree/Fellowship or other similar title to any candidate of any University

Place: ISME Bangalore

Signature of the Student

Date: 10-06-2026

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K. Mohammed Abdulla Khan
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LIST OF ABBREVIATIONS

Abbreviation	Expanded Form
BIS	Bureau of Indian Standards
CCM	Continuous Casting Machine
CRS	Corrosion Resistant Steel
CSA	Cross-Sectional Area
DRI	Direct Reduced Iron
GM	General Manager
HR	Human Resources
IF	Induction Furnace
IS	Indian Standard
ISME	International School of Management Excellence
KPI	Key Performance Indicator
MS	Mild Steel
MSE	Medium Scale Enterprise
MSK	Medvedev-Sponheuer-Karnik (scale)
NHAI	National Highways Authority of India
OpEx	Operational Expenditure
PGDM	Post Graduate Diploma in Management
PSHA	Probabilistic Seismic Hazard Assessment
RM	Raw Material
TMT	Thermo-Mechanically Treated
TS	Tensile Strength
YS	Yield Stress

CHAPTER 1: INTRODUCTION

1.1 Introduction to Internship / Summer Training

Internships serve as a critical bridge between theoretical classroom learning and practical industrial application. The summer internship program at ISME, Bangalore is designed to provide students with hands-on exposure to real-world business operations, enabling them to understand the functioning of organizations, develop professional skills, and apply management concepts in practical settings.

This internship was undertaken at **Hindupur Steel & Alloys (P) Ltd.**, a TMT bar manufacturing plant operating as a franchisee of Kamdhenu Steel, one of India's leading steel brands. The plant is located at APIIC Industrial Park, Gollapuram, Hindupur, Andhra Pradesh. The training spanned from **April 1, 2026 to May 31, 2026**, covering eight weeks of progressive, hands-on exposure to the complete TMT bar production process — from raw material procurement and elemental testing, through smelting and billet casting, to rolling, quenching, final quality control testing, billing, accounts, and financial analysis.

The internship provided end-to-end exposure across multiple departments including:

- Quality Control & Production (Weeks 1-2)
- Billing & Accounts (Weeks 3-4)
- Finance & Cost Analysis (Weeks 5-6)
- Production & Operations Analysis (Weeks 7-8)

1.2 Importance of Internship / Summer Training

The importance of this summer internship can be understood across multiple dimensions:

1. Practical Application of Management Concepts: The internship allowed application of theoretical concepts from operations management, financial accounting, cost analysis, and quality management to real industrial processes.

2. Understanding Cross-Functional Integration: Working across production, quality, billing, accounts, and finance departments provided a holistic understanding of how different functions integrate to achieve organizational goals.

3. Industry Exposure: The steel industry is fundamental to India's infrastructure growth. Understanding its operations, cost structures, quality standards, and market dynamics is invaluable for a management student.

4. Professional Skill Development: The internship developed skills in data analysis, report preparation, documentation, communication, and problem-solving in a real industrial environment.

5. Career Exploration: Exposure to multiple departments helped identify areas of interest and strength for future career specialization.

1.3 Objectives of the Study

The following objectives were set for this summer internship training:

1. **To study the complete production process** of TMT bars at Hindupur Steel & Alloys (P) Ltd., from raw material procurement to finished goods dispatch.
2. **To understand quality control mechanisms** including tensile testing, Ring Test (Etching Test), and Spectro Lab analysis as per IS 1786:2008 standards.
3. **To analyze the billing and accounting processes** including invoice generation, ledger maintenance, and Tally software operations.
4. **To examine the financial operations** including raw material purchase analysis, sales performance tracking, and cost structure analysis.
5. **To assess production efficiency** through yield analysis (RM to Billet, Billet to TMT) and identify areas for improvement.
6. **To conduct profitability and breakeven analysis** to understand the financial health of the organization.
7. **To provide actionable recommendations** based on observations and data analysis.

1.4 Scope and Period of Training

Period of Training: The internship was conducted for a duration of **8 weeks** from **April 1, 2026 to May 31, 2026**.

Scope of Study:

The scope of this internship covered the following activities:

Week	Department	Key Activities
Week 1-2	Quality Control & Production	Tensile testing, Ring Test, TMT production process, raw material testing (Spectro Lab), dispatch loading supervision
Week 3-4	Billing & Accounts	Invoice generation, ledger maintenance, Tally entries, sales & purchase ledgers
Week 5-6	Finance & Cost Analysis	Purchase/sales data analysis, operational costs (electricity, labour), Excel-based reporting
Week 7-8	Production & Operations Analysis	Excel dashboard creation, profit/loss analysis, breakeven analysis, production efficiency tracking

Table 1.1: Internship Schedule - Department-wise Rotation

Geographical Scope: The study was conducted at Hindupur Steel & Alloys (P) Ltd. plant located at APIIC Industrial Park, Gollapuram, Hindupur, Andhra Pradesh.

Functional Scope: The study covered production, quality control, billing, accounts, finance, and operations functions of the organization.

1.5 Meaning of the Concept

TMT Bars (Thermo-Mechanically Treated Bars)

TMT bars are high-strength reinforcement bars used in reinforced cement concrete (RCC) construction. They are manufactured through a specialized process where hot-rolled steel bars are rapidly quenched with water jets, followed by self-tempering using the residual heat from the core. This process creates a unique three-zone microstructure:

- **Tempered Martensite (Outer Ring):** Provides high tensile strength
- **Bainite Transition Zone (Middle):** Provides toughness and shock absorption
- **Ferrite-Pearlite Core (Inner):** Provides ductility and flexibility

Quality Control (QC) in Steel Manufacturing

Quality control in steel manufacturing involves systematic testing of raw materials, in-process materials, and finished products to ensure compliance with BIS standards such as IS 1786:2008. Key tests include:

- Spectro Lab analysis (elemental composition)
- Tensile testing (yield stress, tensile strength, elongation)
- Ring Test (microstructure verification)

Financial Operations

Financial operations in a manufacturing company include procurement of raw materials, tracking of purchases and sales, cost accounting (fixed vs variable costs), power consumption tracking, labour cost management, and profitability analysis.

Breakeven Analysis

Breakeven analysis determines the level of sales volume at which total revenue equals total costs (fixed + variable), resulting in zero profit or loss. For Hindupur Steels, the breakeven quantity was calculated as 237 tons per day.

1.6 Rationale for Choosing the Topic

The topic "**Analysis of Purchase of Raw Materials, Production, Sales and Financial Operations**" was chosen for the following reasons:

1. **End-to-End Exposure:** The internship provided rotation across multiple departments, allowing a comprehensive view of the entire value chain from raw material purchase to finished goods sales.
2. **Industry Relevance:** The steel industry is a core sector of the Indian economy, and understanding its operations is highly relevant for a management student.
3. **Data Availability:** The organization maintained detailed records of production, quality, purchase, sales, and financial data, enabling meaningful analysis.

4. **Practical Learning Opportunity:** The topic allowed integration of concepts from operations management, financial accounting, cost management, and quality management.
5. **Value Addition to Organization:** The analysis conducted (5-day profitability dashboard, breakeven analysis, efficiency metrics) provided actionable insights to the management.

1.7 Implementation Strategy of the Concept in Present Study

The implementation strategy for this study involved:

Phase 1: Observation & Learning (Weeks 1-4)

- Direct observation of production processes
- Hands-on training in quality control lab
- Learning billing and accounting procedures

Phase 2: Data Collection (Weeks 5-6)

- Collection of purchase, sales, production, and operational data
- Understanding cost structures (power, labour, raw materials)

Phase 3: Analysis & Reporting (Weeks 7-8)

- Creation of Excel dashboards
- Calculation of KPIs, yields, and breakeven point
- Preparation of final report with findings and recommendations

CHAPTER 2: COMPANY PROFILE

2.1 Introduction to Company

Hindupur Steel & Alloys (P) Ltd. is a private limited company engaged in the manufacturing of MS Billets and Kamdhenu NXT TMT Bars. The company was incorporated in 2009 and operates as a franchisee of Kamdhenu Ltd, one of India's most recognized steel brands. The company's manufacturing plant is strategically located at APIIC Industrial Park, Gollapuram, Hindupur, Andhra Pradesh, benefiting from proximity to raw material sources and excellent road connectivity via NH 44 (Bengaluru-Hyderabad highway).

Company Details:

- **Complete Address:** Plot No. 29, APIIC Industrial Park, Gollapuram, Hindupur Mandal, Andhra Pradesh - 515211
- **Telephone:** +91 9505681005
- **Email:** hindupursteelhup@gmail.com

- **Website:** (Operating under Kamdhenu brand)

Vision:

"To be a trusted and leading manufacturer of high-quality steel products in India, contributing to the nation's infrastructure growth through innovation, quality, and sustainable manufacturing practices."

Mission:

- To manufacture and deliver world-class MS Billets and Kamdhenu NXT TMT Bars that meet highest standards of quality and durability
- To build long-term relationships with customers based on trust, transparency, and consistent product performance
- To optimize manufacturing processes for cost efficiency, energy conservation, and minimal environmental impact
- To contribute to socio-economic development of Hindupur region through employment and community initiatives

Objectives:

- Expand annual production capacity of MS Billets and TMT Bars
- Maintain ISO-aligned quality standards across all manufacturing stages
- Strengthen market presence across Andhra Pradesh, Karnataka, and neighboring states
- Invest in technology upgrades to improve operational efficiency
- Comply with all statutory, environmental, and labor regulations

2.2 Area of Operation

Hindupur Steel & Alloys (P) Ltd. operates at a **national level** within India, with concentrated activity in southern and central Indian states.

Registered Office: Bangalore, Karnataka

Plant Location: APIIC Industrial Park, Gollapuram, Hindupur, Andhra Pradesh

Geographical Areas of Operation:

- Andhra Pradesh (primary market)
- Karnataka (secondary market)
- Tamil Nadu (emerging market)
- Telangana (emerging market)

The plant's strategic location near NH 44 enables efficient dispatch of finished goods to customers across South India.

2.3 Nature of the Organization and its Business

Type of Industry: Manufacturing (Basic Iron & Steel)

Legal Status: Private Limited Company (Non-Government)

NIC Code: 271 – Manufacture of Basic Iron & Steel

Business Type: Business-to-Business (B2B) Manufacturing

Specific Functional Areas: Production, Quality Control, Finance & Accounts, Human Resources, Marketing & Sales

The company is engaged exclusively in the manufacturing of two core steel products:

1. MS Billets (Mild Steel Billets) - semi-finished products for re-rolling mills
2. Kamdhenu NXT TMT Bars - finished reinforcement bars for construction

2.4 Product or Service Profile

Table 2.1: Basic Company Information

Parameter	Details
Company Name	Hindupur Steel & Alloys Pvt Ltd
Incorporation Year	2009
Type	Private Limited Company (Non-Government)
Head Office	Bangalore, Karnataka
Plant Location	APIIC Industrial Park, Gollapuram, Hindupur, Andhra Pradesh
NIC Code	271 – Manufacture of Basic Iron & Steel
Core Products	MS Billets & Kamdhenu NXT TMT Bars
Authorized Capital	₹30-35 Crore
Paid-up Capital	₹20-25 Crore

Table 2.2: Product Profile - MS Billets

Parameter	Details
Product Name	MS Billets (Mild Steel Billets)
Grade	IS 2830 / IS 2831 compliant
Cross Section	100mm x 100mm to 130mm x 130mm
Length	3 to 12 metres (customizable)
Application	Re-rolling mills – conversion into TMT bars, angles, channels, rods

Key Property	High cleanliness, low sulphur/phosphorus content, uniform microstructure
Market	Local and inter-state re-rollers; B2B sales

Table 2.3: Product Profile - Kamdhenu NXT TMT Bars

Parameter	Details
Brand	Kamdhenu NXT
Standard	IS 1786:2008 (BIS Certified)
Grades	Fe-415, Fe-500, Fe-500D
Sizes Available	8mm, 10mm, 12mm, 16mm, 20mm, 25mm, 32mm
Surface	Ribbed / Deformed (for superior concrete bonding)
Key Features	High ductility, corrosion resistance, earthquake resistance, weldability
Applications	Residential buildings, bridges, dams, flyovers, industrial structures

Product Advantages:

- **Superior Strength:** Meets and exceeds BIS standards for yield strength and tensile strength
- **Ductility:** Soft ferrite-pearlite inner core allows bending without cracking – ideal for earthquake-prone regions
- **Corrosion Resistance:** Tempered martensite outer rim provides natural resistance to oxidation
- **Weldability:** Low carbon equivalent ensures easy and reliable welding on construction sites
- **Brand Trust:** Kamdhenu NXT brand is widely recognized by builders, contractors, and government agencies

2.5 Size of the Organization

Table 2.4: Manpower and Turnover Details

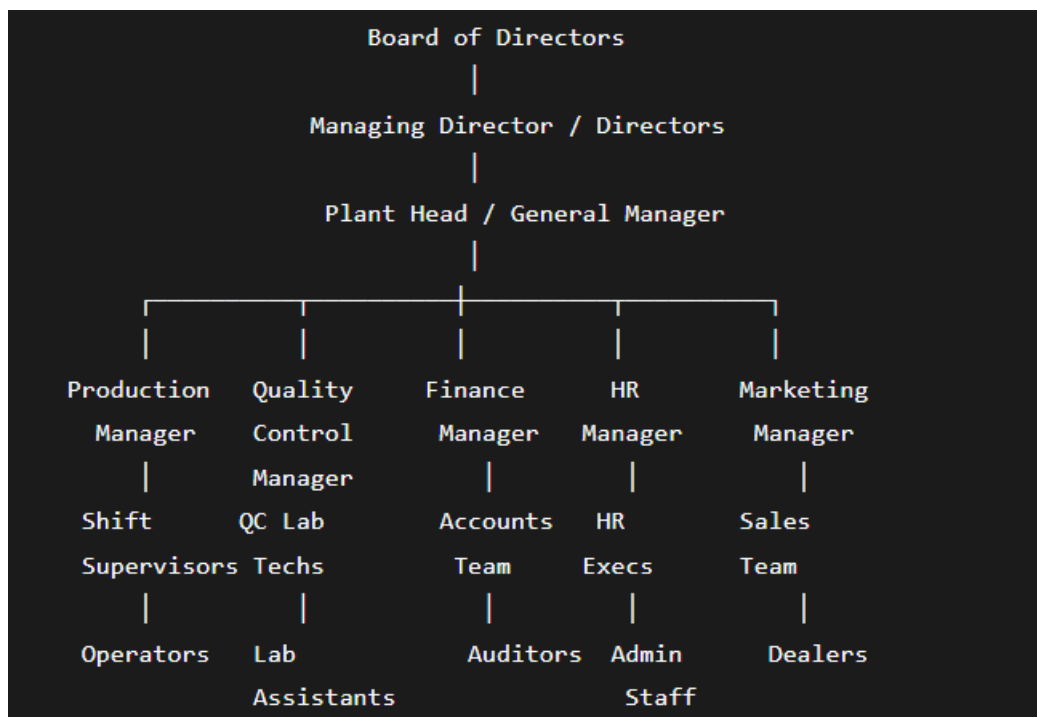
Parameter	Details
Employee Strength	~250-300 employees (including permanent and contractual)
Annual Turnover	₹750 – 1,000 Crore (approx.)
Authorized Capital	₹20-25 Crore
Paid-up Capital	₹30-35 Crore
Plant Area	APIIC Industrial Park, Hindupur (dedicated industrial zone)
Production Capacity	Approx. 1,00,000 MT per annum (MS Billets + TMT Bars combined)
Current Production	~600-700 Metric Tons per Day

Number of Shifts	3 shifts (24-hour operations)
Classification	Medium-Scale Enterprise (MSE) under MSME definition

2.6 Organization Structure of the Company

Hindupur Steel & Alloys (P) Ltd. follows a **Functional Organizational Structure**. In this structure, employees are grouped according to their specialized functions such as Production, Quality Control, Finance, HR, and Marketing. This enables departmental expertise, clear reporting lines, and efficient resource utilization in a manufacturing environment.

Figure 2.1: Organisation Structure Chart



Key Functional Departments:

- 1. Production Department:** Manages induction furnace operations, billet casting, and TMT rolling mill
- 2. Quality Control (QC) Department:** Conducts incoming raw material testing, in-process checks, and finished goods testing per BIS standards
- 3. Finance & Accounts:** Handles budgeting, cost control, payroll, statutory compliance, and banking relationships
- 4. Human Resources (HR):** Manages recruitment, training, performance appraisal, welfare, and legal compliance
- 5. Marketing & Sales:** Handles customer acquisition, dealer network management, order processing, and dispatch coordination

2.7 Competitor Information

The Indian steel industry is served by large integrated steel plants as well as numerous small and medium secondary producers. Key competitors in the TMT bar and billet segment include:

Company	Key Products	Remarks
Tata Steel	TMT Bars, Structurals, Wire Rods	Integrated; Pan-India presence
JSW Steel	TMT Bars, HR Coils, Billets	Largest private steel producer
SAIL	TMT Bars, Billets, Plates	Government-owned; major long products
Jindal Steel & Power (JSPL)	TMT Bars, Billets, Rail	Strong in Central & Eastern India
Vizag Steel (RINL)	TMT Bars, Wire Rods	Located in AP; key regional competitor
Kamdhenu Ltd (other franchisees)	Kamdhenu TMT Bars	Direct brand-level competition

Figure 2.2: Raw Material Purchase Mix

(A pie chart showing: Sponge Iron 43%, Ingots & Billets 36%, Turning & Boring Scrap 21% - based on Table 2.1 from your final report)

Hindupur Steel & Alloys competes primarily at the regional level in Andhra Pradesh and Karnataka. Its competitive advantage lies in:

- Kamdhenu NXT brand recognition
- Proximity to construction markets
- Cost efficiency due to scrap-based induction furnace production

CHAPTER 3: INDUSTRY PROFILE

3.1 Brief Profile of the Industry

The Indian steel industry is one of the most significant sectors of the national economy, contributing substantially to GDP, employment, and infrastructure development. India has emerged as the world's **second-largest steel producer**, surpassing Japan, and trails only China in global steel output.

Table 3.1: Indian Steel Industry Overview

Parameter	Value
India's Steel Production (2023-24)	~144 Million Tonnes (MT)
Global Rank	2nd Largest Steel Producer
Steel Consumption (2023-24)	~136 MT
Per Capita Steel Consumption	~97 kg (vs. global average ~228 kg)
Industry Contribution to GDP	Approx. 2% directly; 6-7% indirectly
Employment	Over 6 lakh direct; 20+ lakh indirect
Target (National Steel Policy 2017)	300 million MT by 2030-31

India's steel industry benefits from:

- Large domestic iron ore reserves
- Growing domestic market
- Strong government push through infrastructure spending
- Availability of skilled and semi-skilled labour
- Competitive manufacturing costs

The **TMT bar segment**, in which Hindupur Steel operates, accounts for a significant share of long products consumption driven by construction activity across residential, commercial, and infrastructure sectors.

3.2 Present Industry Status

The Indian steel industry is witnessing robust growth driven by the government's emphasis on infrastructure development under programs like:

- PM Gati Shakti (National Master Plan for multi-modal connectivity)
- Smart Cities Mission
- Pradhan Mantri Awas Yojana (PMAY)
- Bharatmala Pariyojana (highway development)
- Sagarmala (port-led development)

Key Industry Trends as of 2024-26:

1. **Demand Growth:** Construction and infrastructure sectors are driving consumption of TMT bars at record levels. The housing sector alone accounts for approximately 60% of TMT bar consumption.
2. **Raw Material Volatility:** Prices of scrap steel, iron ore, and coking coal continue to fluctuate due to global supply chain disruptions and geopolitical factors.

3. **Energy Costs:** Electricity remains a major cost component for induction furnace producers like Hindupur Steel. Rising power tariffs in various states pose a challenge to profitability.
4. **Environmental Compliance:** Increasing pressure from the National Green Tribunal (NGT) and state pollution control boards for emissions reduction and effluent treatment.
5. **BIS Certification:** Mandatory BIS marking for TMT bars (IS 1786) under the Steel & Steel Products (Quality Control) Order has raised the bar for quality compliance and eliminated many sub-standard producers.
6. **Capacity Utilization:** The secondary steel sector (induction furnace route) operates at approximately 70-75% capacity utilization nationally.
7. **Consolidation Trend:** Larger players are acquiring smaller units to achieve economies of scale and expand market reach.

3.3 Future Prospects of the Industry

The outlook for the Indian steel industry, and specifically for TMT bar producers, is highly positive over the medium to long term:

1. **National Steel Policy 2017 Target:** India aims to achieve 300 MT steel production capacity by 2030-31, implying massive capacity addition across the value chain. This represents a doubling of current production levels.
2. **Infrastructure Pipeline:** Government capital expenditure on roads, railways, metro rail, ports, and affordable housing is expected to sustain high TMT bar demand for the next decade. The National Infrastructure Pipeline (NIP) has identified over ₹111 lakh crore worth of projects.
3. **Urbanization:** India's urbanization rate is projected to increase from ~36% (2023) to ~50% by 2050, driving residential construction demand. Each 1% increase in urbanization adds approximately 10-12 MT of steel demand.
4. **Per Capita Consumption Growth:** India's per capita steel consumption of ~97 kg is well below the global average of ~228 kg, leaving significant room for growth as incomes rise. Even reaching the global average would triple domestic demand.
5. **Export Opportunity:** India is increasingly exporting steel to neighbouring countries and Africa; secondary producers can benefit from export demand for billets and TMT bars.
6. **Make in India:** Government incentives for domestic manufacturing, including Production Linked Incentive (PLI) schemes for specialty steel, will benefit quality-focused producers.
7. **Green Steel Transition:** Future regulations may favour electric arc/induction furnace producers (like Hindupur Steel) over blast furnace producers due to lower carbon emissions per tonne. The global push for decarbonization presents both challenges and opportunities.

8. Seismic Safety Awareness: Growing awareness of earthquake-resistant construction, particularly after recent tremors in Delhi-NCR and other regions, is driving demand for higher-grade TMT bars (Fe 500D, Fe 550D).

CHAPTER 4: ORGANISATION OVERVIEW

4.1 Various Departments of the Organization

Hindupur Steel & Alloys (P) Ltd. is organized into the following key departments:

1. Production Department
2. Quality Control (QC) Department
3. Finance & Accounts Department
4. Human Resources (HR) Department
5. Marketing & Sales Department

4.2 Functioning of Various Departments

4.2.1 Finance Department

Introduction: The Finance Department at Hindupur Steel & Alloys (P) Ltd. is responsible for managing the company's financial resources, maintaining books of accounts, ensuring statutory compliance, and supporting decision-making through financial analysis and reporting. The department works closely with production, purchase, and sales teams to track costs, manage working capital, and optimize profitability.

Functions of Finance Department:

1. **Financial Accounting & Record Keeping:** Maintaining day-to-day transaction records, ledgers, and financial statements
2. **Purchase & Sales Accounting:** Recording raw material purchases, tracking supplier payments, and recording sales invoices
3. **Cost Management:** Tracking production costs (raw materials, power, labour), calculating cost per ton, and identifying cost reduction opportunities
4. **Statutory Compliance:** GST filing, TDS deduction and payment, income tax compliance, and other regulatory requirements
5. **Payroll Management:** Salary calculation, disbursement, and statutory deductions (PF, ESI)
6. **Banking & Treasury:** Managing bank accounts, fund transfers, and cash flow
7. **Financial Reporting:** Preparing monthly profit & loss statements, balance sheets, and management information system (MIS) reports

8. **Budgeting & Variance Analysis:** Preparing annual budgets and comparing actual performance against budget

Figure 4.1: Finance Department Structure

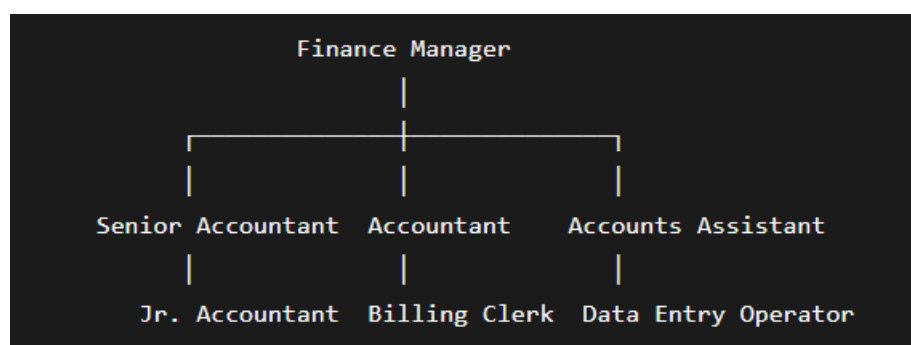


Table 4.1: Finance Department - Roles and Responsibilities

Level	Roles	Responsibilities
Finance Manager	Department Head	Overall financial management, statutory compliance, banking relationships, financial planning, cost analysis
Senior Accountant	Team Lead	Finalization of accounts, GST & TDS returns, payroll processing, MIS reporting
Accountant	Execution	Day-to-day accounting, ledger maintenance, invoice verification, Tally entries
Accounts Assistant / Billing Clerk	Support	Invoice generation, data entry, filing, document organization

4.2.2 Human Resources Department

Introduction: The Human Resources Department at Hindupur Steel & Alloys (P) Ltd. manages the company's workforce of approximately 250-300 employees, including permanent staff and contractual workers. The department focuses on recruitment, training, employee welfare, compliance with labour laws, and maintaining a positive work environment.

Functions of HR Department:

1. **Recruitment & Selection:** Identifying manpower requirements, sourcing candidates, conducting interviews, and onboarding new employees
2. **Training & Development:** Organizing technical training for operators, safety training, and soft skills development
3. **Attendance & Leave Management:** Tracking daily attendance, managing leave records, and coordinating with payroll

4. **Statutory Compliance:** Compliance with Factories Act, PF, ESI, Bonus Act, Gratuity Act, and other labour laws
5. **Employee Welfare:** Managing canteen facilities, safety equipment, health check-ups, and employee grievance handling
6. **Performance Management:** Conducting annual performance appraisals and managing increments
7. **Contract Labour Management:** Managing contractual workforce, vendor coordination, and compliance for contract labour

Figure 4.2: HR Department Structure

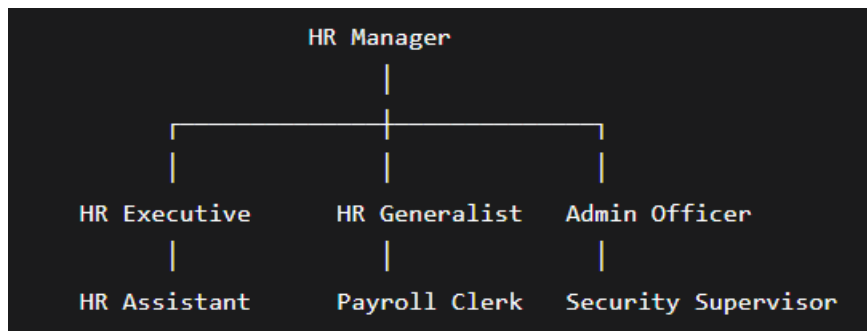


Table 4.2: HR Department - Roles and Responsibilities

Level	Roles	Responsibilities
HR Manager	Department Head	HR strategy, policy formulation, statutory compliance, employee relations, performance management
HR Executive	Recruitment & Compliance	Recruitment, onboarding, training coordination, PF/ESI compliance
HR Generalist	Operations	Attendance management, employee records, grievance handling, welfare activities
Admin Officer	Administration	Facility management, security supervision, canteen management, vendor coordination

4.2.3 Marketing Department

Introduction: The Marketing & Sales Department at Hindupur Steel & Alloys (P) Ltd. is responsible for promoting and selling Kamdhenu NXT TMT bars to dealers, contractors, and institutional customers. The department also manages customer relationships, order processing, and dispatch coordination.

Functions of Marketing Department:

1. **Dealer Management:** Onboarding and managing dealers across Andhra Pradesh, Karnataka, and other markets

2. **Order Processing:** Receiving customer orders, confirming product availability, coordinating with production and dispatch
3. **Sales Tracking:** Recording daily sales (quantity and value), tracking brand-wise sales (NXT vs KAY2)
4. **Customer Relationship Management:** Managing relationships with contractors, builders, and institutional customers
5. **Market Intelligence:** Tracking competitor prices, market trends, and customer preferences
6. **Dispatch Coordination:** Coordinating with production and logistics teams for timely delivery
7. **Brand Promotion:** Promoting the Kamdhenu NXT brand through dealer meets, site visits, and promotional materials

Figure 4.3: Marketing Department Structure

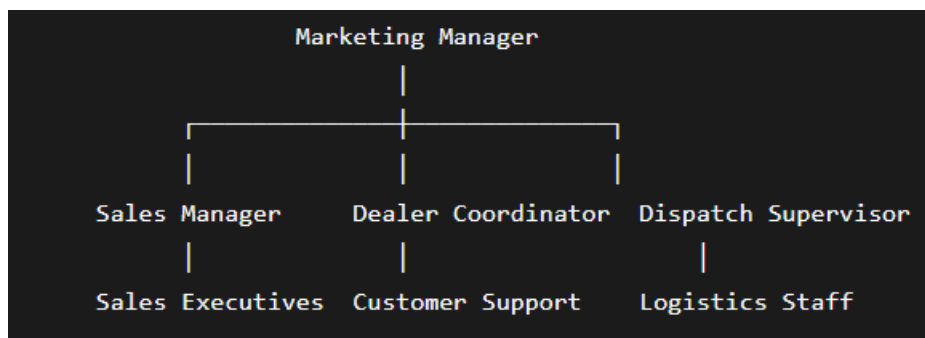


Table 4.3: Marketing Department - Roles and Responsibilities

Level	Roles	Responsibilities
Marketing Manager	Department Head	Sales strategy, dealer network development, pricing decisions, team management
Sales Manager	Sales Operations	Order processing, sales target achievement, customer relationship management
Dealer Coordinator	Dealer Management	Dealer onboarding, order coordination, payment follow-up
Dispatch Supervisor	Logistics	Loading supervision, vehicle coordination, delivery documentation

CHAPTER 5: INTERNSHIP WORK OVERVIEW

5.1 Tasks Performed

The internship involved rotation across four major functional areas over eight weeks. The tasks performed in each period are detailed below:

Weeks 1-2: Quality Control & Production Department

Tasks Performed:

1. Tensile Testing (Grade Testing Laboratory):

- Collected rebar samples (8mm, 12mm, 20mm) every 30 minutes from production floor
- Measured sample weight, length, and cross-sectional area
- Operated tensile testing machine to determine yield load, tensile load, yield stress, tensile strength, and elongation percentage
- Recorded test results and compared against IS 1786:2008 standards

2. Ring Test (Etching Test) Study:

- Prepared cross-section samples of TMT bars
- Polished samples to mirror finish
- Applied 5-10% Nital solution (nitric acid in ethanol) for 15-30 seconds
- Observed three-zone microstructure under magnification
- Determined pass/fail based on continuity of martensite ring

3. Raw Material Testing (Spectro Lab):

- Prepared lollipop-shaped samples from incoming raw materials (scrap, sponge iron, pig iron)
- Ground samples to flat, shining surface
- Conducted Spectro Lab analysis (5-10 seconds per test)
- Compared elemental composition against required standards

4. TMT Production Process Observation:

- Observed induction furnace operations (melt temperature ~1380°C)
- Understood billet casting through CCM (Continuous Casting Machine)
- Observed rolling mill operations
- Studied quenching and self-tempering process

5. Dispatch Loading Supervision:

- Supervised loading of TMT bars of various sizes onto delivery vehicles
- Verified customer orders against loaded quantities

Weeks 3-4: Billing & Accounts Department

Tasks Performed:

1. Invoice Generation:

- Learned invoice preparation process for TMT bar sales
- Ensured accuracy of customer details, product specifications, quantity, and pricing

2. Ledger Maintenance:

- Maintained sales and purchase ledgers
- Recorded daily transactions systematically

3. Tally Software Operations:

- Entered invoices into Tally software
- Verified entries after recording
- Understood basic accounting workflows in Tally

Weeks 5-6: Finance & Cost Analysis Department

Tasks Performed:

1. Purchase Data Analysis:

- Collected and organized raw material purchase data (turning & boring scrap, ingots & billets, sponge iron)
- Analyzed purchase quantity, value, and price per kg
- Created material-wise purchase summary

2. Sales Data Analysis:

- Collected daily TMT sales data (quantity and value)
- Analyzed brand-wise sales (Kamdhenu NXT vs KAY2)
- Calculated average selling price per ton

3. Operational Cost Analysis:

- Tracked daily power consumption for furnace and rolling mill
- Calculated power cost at ₹7 per unit
- Understood labour cost structure

4. Excel-Based Reporting:

- Created Excel sheets for purchase, sales, and cost tracking
- Prepared summary tables and charts

Weeks 7-8: Production & Operations Analysis

Tasks Performed:

1. Profitability Analysis:

- Prepared 5-day Profit & Loss statement (May 1-5, 2026)
- Calculated gross profit and net profit
- Determined net profit margin (11.9% over 5 days)

2. Production Efficiency Analysis:

- Calculated raw material to billet conversion yield (83.9%)
- Calculated billet to TMT conversion yield (95.8%)
- Identified wastage quantities and percentages

3. Breakeven Analysis:

- Calculated daily fixed operational cost (₹43.67 Lakhs)
- Determined contribution margin per ton (₹18,414)
- Calculated breakeven quantity (237 tons per day)
- Calculated breakeven revenue (₹1.26 Crore per day)

4. Excel Dashboard Creation:

- Created interactive Excel dashboard with profit analysis sheets
- Added production efficiency charts and operational cost breakdowns
- Enabled identification of loss-making days

5.2 Process Followed

Process 1: TMT Bar Manufacturing Process

The complete TMT bar manufacturing process at Hindupur Steel & Alloys (P) Ltd. involves the following stages:

Stage 1: Raw Material Procurement

- Collection of iron scrap, sponge iron, pig iron, and gypsum
- Storage in scrap yard

Stage 2: Raw Material Testing (Spectro Lab)

- Sample preparation (lollipop-shaped, polished flat)

- Spectro Lab analysis (5-10 seconds)
- Decision: Accept → Furnace OR Reject → Return to supplier

Stage 3: Induction Furnace Melting

- Charge composition: ~6 tons scrap + ~14 tons pellets/DRI (total ~20 tons)
- Melting temperature: ~1380°C
- Power consumption: ~722 kWh per tonne
- Cycle time: ~3 hours per 18-tonne batch

Stage 4: Continuous Casting (CCM Unit)

- Molten steel poured into continuous casting machine
- Solidification into billets
- Output: ~18 tonnes of billets per heat (~11 billets of ~840 kg each)

Stage 5: Rolling Mill

- Billets reheated and passed through rolling stands
- Reduced to required diameter (8mm to 32mm)

Stage 6: Quenching (Thermo-Mechanical Treatment)

- Hot bar passed through water jets
- Outer surface quenched rapidly → martensite formation
- Core remains hot → ferrite-pearlite

Stage 7: Self-Tempering (Air Cooling)

- Residual heat from core tempers outer martensite
- Three-zone microstructure formed

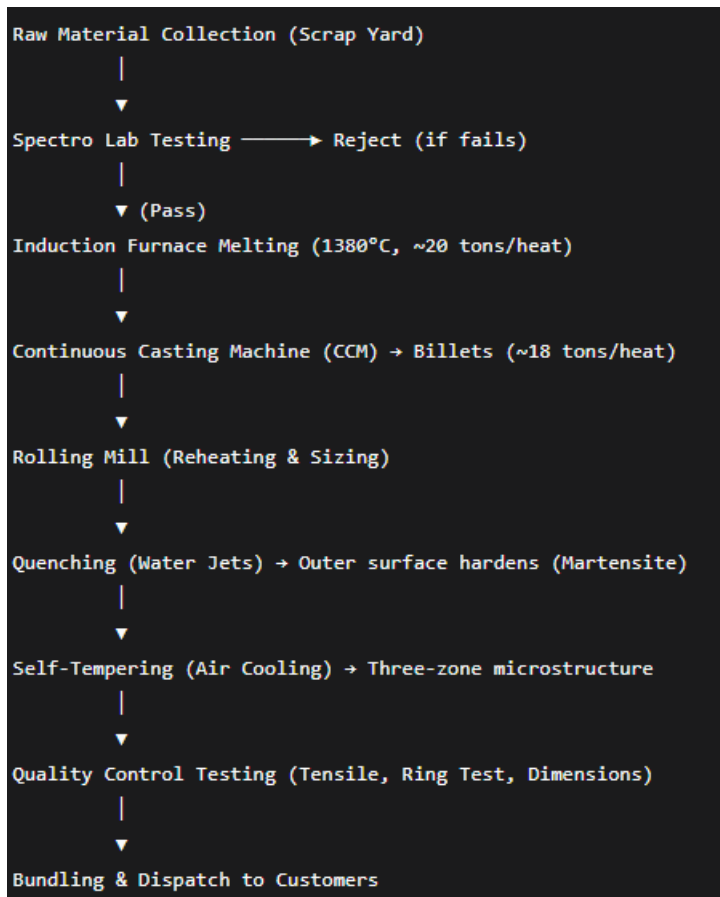
Stage 8: Quality Control Testing

- Tensile testing every 30 minutes
- Ring Test (Etching Test) as per IS 1786
- Dimensional checks (weight per metre within tolerance)

Stage 9: Dispatch

- Bundling as per size (8mm: 10 rods/bundle, 20mm: 2 rods/bundle, etc.)
- Loading onto vehicles
- Delivery to customers

Figure 5.1: Process Flow - Raw Material to Finished TMT Bars



Process 2: Tensile Testing Procedure (Followed Every 30 Minutes)

Step 1: Sample collected from production line at specified time

Step 2: Sample weighed and length measured

Step 3: Cross-sectional area calculated

Step 4: Sample placed in tensile testing machine

Step 5: Tensile load applied gradually

Step 6: Yield load (elastic limit) recorded

Step 7: Ultimate tensile load (break point) recorded

Step 8: Calculations performed:

- Yield Stress = Yield Load / CSA
- Tensile Strength = Tensile Load / CSA
- Elongation % = (Final length - Original length) / Original length × 100

Step 9: Results compared against IS 1786 standards

Step 10: Pass/Fail decision recorded

Process 3: Ring Test (Etching Test) Procedure

Step 1: Cross-section cut from TMT bar sample

Step 2: Cut surface polished to mirror finish

Step 3: 5-10% Nital solution prepared (nitric acid in ethanol)

Step 4: Polished surface immersed in Nital for 15-30 seconds

Step 5: Sample removed and washed

Step 6: Microstructure observed under magnification

Step 7: Three zones identified:

- Tempered Martensite (dark/black outer ring)
 - Transition zone (medium/brown)
 - Ferrite-Pearlite core (light/white)
- Step 8:** PASS if dark ring is continuous and unbroken
Step 9: FAIL if dark ring has breaks or unevenness

5.3 Results Achieved

Result 1: 5-Day Profitability Analysis (May 1-5, 2026)

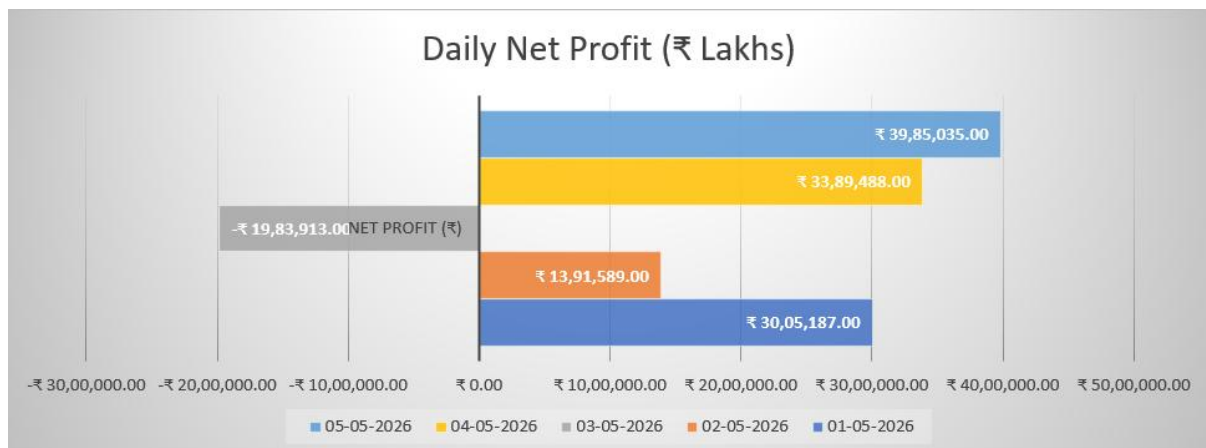
Based on the data collected and analyzed during Weeks 7-8, the following results were achieved:

Table 5.1: Daily Profit & Loss Statement

Date	Sales Revenue (₹)	RM Cost (₹)	Gross Profit (₹)	OpEx (Fixed) (₹)	Net Profit (₹)	Net Margin %
01-May-26	1,94,92,154	1,27,46,000	67,46,154	43,66,667	30,05,187	15.4%
02-May-26	1,48,30,556	96,98,000	51,32,556	43,66,667	13,91,589	9.4%
03-May-26	50,79,054	33,22,000	17,57,054	43,66,667	(19,83,913)	-39.1%
04-May-26	2,06,16,455	1,34,86,000	71,30,455	43,66,667	33,89,488	16.4%
05-May-26	2,23,48,002	1,46,22,000	77,26,002	43,66,667	39,85,035	17.8%
Total	8,23,66,221	5,38,74,000	2,84,92,221	2,18,33,335	98,02,558	11.9%

Table 5.2: Key Performance Indicators (KPIs)

KPI	Value
Total Sales Revenue	₹8.24 Crore
Total Sales Quantity	1,548.08 Tons
Total Net Profit	₹98.03 Lakhs
Net Profit Margin	11.9%
Average Selling Price	₹53,205 per Ton
Average Variable Cost (RM)	₹34,791 per Ton
Daily Breakeven Quantity	~237 Tons
Daily Breakeven Revenue	~₹1.26 Crore

Figure 5.2: Daily Net Profit / Loss Trend

Key Finding: The company is fundamentally profitable on average days, but Day 3 (May 3) saw a significant net loss of ₹19.83 Lakhs due to low sales volume (only 95 tons) falling below the breakeven quantity of 237 tons.

Result 2: Production Efficiency Analysis

Table 5.3: Production Efficiency - Yield Analysis

Process	Input	Output	Yield	Wastage
Raw Material → Billet	1,775.20 Tons	1,490 Tons	83.9%	285.20 Tons (16.1%)
Billet → TMT	1,840.61 Tons	1,744.55 Tons	95.8%	96.06 Tons (4.2%)

Key Finding: The rolling mill conversion (95.8% yield) is highly efficient and within industry standards. The primary melting process (83.9% yield) has room for improvement.

Result 3: Power Consumption Analysis

Table 5.4: Daily Power Consumption - Furnace vs Rolling Mill

Parameter	Total Units	Total Cost (₹ @ ₹7/unit)
Furnace (5 days)	11,94,600	1,19,84,000
Rolling Mill (5 days)	2,35,900	16,65,800
Combined	14,30,500	1,36,49,800

Key Finding: Power cost is a major operational expense, with the furnace consuming approximately 5 times more electricity than the rolling mill.

Result 4: Brand-wise Sales Analysis

Table 5.5: Brand-wise Sales Breakdown (NXT vs KAY2)

Brand	Sales Quantity (Kgs)	Sales Value (₹)	Avg. Price (₹/Kg)
Kamdhenu NXT	13,30,390	7,12,36,381	53.55
KAY2	2,17,690	1,11,29,840	51.13
Total	15,48,080	8,23,66,221	53.20

Key Finding: Kamdhenu NXT (premium brand) accounts for 86% of sales quantity and achieves a ₹2.42/kg higher price realization compared to the KAY2 economy brand.

Figure 5.3: Daily Sales Quantity vs. Breakeven

(Line chart showing: Day 1: 364 tons, Day 2: 277 tons, Day 3: 95 tons (below 237 breakeven), Day 4: 388 tons, Day 5: 424 tons)

Key Finding: Only Day 3 fell below the breakeven quantity of 237 tons, resulting in the only loss-making day.

5.4 Conclusions, Learnings & Suggestions

Conclusions

Based on the 8-week internship and the 5-day detailed analysis, the following conclusions are drawn:

- Financial Performance:** Hindupur Steel & Alloys (P) Ltd. is fundamentally profitable with an 11.9% net margin over the 5-day analysis period. The company generated ₹8.24 Crore in sales revenue and ₹98 Lakhs in net profit.

2. **Volume Sensitivity:** The company has high operational leverage. A single low-volume day (95 tons on May 3) erased profits from two profitable days. Maintaining daily sales above 240 tons is critical.
3. **Production Efficiency:** The rolling mill conversion (95.8% yield) is excellent. The primary melting process (83.9% yield) has improvement potential of approximately 5-6%.
4. **Quality Standards:** The company maintains robust quality control through Spectro Lab (raw material), tensile testing every 30 minutes, and Ring Test (microstructure verification). All tests are conducted as per IS 1786:2008 standards.
5. **Brand Strategy:** Kamdhenu NXT (premium brand) dominates sales (86% of volume) and achieves better price realization (₹2.42/kg higher than KAY2). This justifies the franchise model.
6. **Cost Structure:** Power cost (₹1.37 Crore over 5 days) is the largest operational expense, with the furnace consuming 83% of total power.

Learnings

The internship provided the following key learnings:

Technical Learnings:

1. **TMT Bar Manufacturing:** Understood the complete production process from raw material to finished TMT bars, including induction furnace melting, continuous casting, rolling, quenching, and self-tempering.
2. **Quality Control:** Learned tensile testing procedures, Ring Test (Etching Test) for microstructure verification, and Spectro Lab analysis for elemental composition.
3. **IS Standards:** Gained working knowledge of IS 1786:2008 (TMT bar specifications), IS 1893:2025 (seismic zones), and tolerance ranges for different bar sizes.
4. **Industry Economics:** Understood cost structures including power consumption (₹7/unit, 722 kWh/tonne), raw material pricing, labour costs, and breakeven calculations.

Software & Analytical Learnings:

1. **Tally Software:** Basic proficiency in invoice entry, ledger maintenance, and transaction recording
2. **Excel Dashboard Creation:** Ability to create P&L statements, breakeven analysis, production efficiency trackers, and data visualizations
3. **Data Analysis:** Practical experience in collecting, organizing, and analyzing operational data

Soft Skills Developed:

1. **Communication:** Interacting with department heads, supervisors, and workers across different functions
2. **Documentation:** Maintaining daily reports, fortnightly reports, and final analysis
3. **Problem-Solving:** Identifying the cause of the Day 3 loss (low sales volume) and calculating breakeven threshold
4. **Professionalism:** Understanding workplace culture, reporting structure, and professional conduct

Suggestions

Based on observations and analysis, the following suggestions are offered to the management:

Suggestion 1: Stabilize Daily Sales Volume

- **Observation:** Day 3 loss was caused by sales volume of only 95 tons vs breakeven of 237 tons
- **Recommendation:** Implement daily sales target of 240+ tons; consider promotional pricing or dealer incentives on low-volume days
- **Expected Benefit:** Elimination of loss-making days

Suggestion 2: Optimize Labour Cost Structure

- **Observation:** Labour cost of ₹43.67 Lakhs over 5 days (₹8.73 Lakhs/day) is treated as fixed cost
- **Recommendation:** Shift labour partially to variable pay model (performance-linked incentives) or optimize shift scheduling based on production volume
- **Expected Benefit:** Lower daily breakeven point

Suggestion 3: Align Production with Sales Orders

- **Observation:** Billet production (2,031 tons) exceeded TMT sales (1,548 tons) → inventory buildup
- **Recommendation:** Implement production planning based on actual sales orders rather than running at full capacity continuously
- **Expected Benefit:** Reduced inventory carrying cost

Suggestion 4: Investigate Root Cause of May 3 Sales Drop

- **Observation:** Sales dropped from 277 tons (Day 2) to 95 tons (Day 3) without clear reason
- **Recommendation:** Analyze whether this was due to demand issue (no orders), supply issue (production problem), or distribution issue (vehicle availability)

- **Expected Benefit:** Prevention of recurrence

Suggestion 5: Improve Primary Melting Yield

- **Observation:** Raw material to billet yield is 83.9%, leaving room for improvement
- **Recommendation:** Analyze slag composition to identify metal loss; train operators on optimal charge mix
- **Expected Benefit:** Additional 5-6% yield would add ~₹50-60 Lakhs annual profit

Suggestion 6: Consider Solar Power Investment

- **Observation:** Power cost is ₹1.37 Crore over 5 days (₹8.2 Crore/month projected)
- **Recommendation:** Evaluate feasibility of rooftop solar plant to reduce grid power dependency
- **Expected Benefit:** Long-term reduction in variable costs and carbon footprint

FUTURE POSSIBILITIES

Based on the internship experience and analysis conducted, the following future possibilities are identified for Hindupur Steel & Alloys (P) Ltd.:

1. Expansion of Production Capacity

With India's steel demand projected to reach 300 MT by 2030-31, the company can explore expanding annual capacity from current 1,00,000 MT to 1,50,000-2,00,000 MT over 3-5 years.

2. Diversification into Higher-Grade Products

- **Fe 550D and Fe 600 Grades:** Growing demand for high-ductility TMT bars in seismic zones (North India) and infrastructure projects
- **Corrosion-Resistant Steel (CRS):** Growing demand in coastal markets (Tamil Nadu, Kerala, Andhra Pradesh)

3. Forward Integration into Construction

The company could consider establishing a small construction division or partnering with developers to create a "Kamdhenu-certified builder" program, similar to brand endorsement models.

4. Energy Efficiency Initiatives

- Solar power plant installation (reducing grid dependency)
- Waste heat recovery from induction furnace (preheating scrap or generating steam)
- Energy-efficient motors and variable frequency drives (VFDs) for rolling mill

5. Digital Transformation

- **ERP Implementation:** Integrating production, quality, inventory, sales, and finance on a single platform
- **Real-time Dashboard:** Management dashboard showing daily production, sales, profitability vs breakeven
- **Automated Quality Reporting:** Digital capture of tensile test results with automatic pass/fail alerts

6. Export Opportunities

With India's growing steel exports, the company could explore exporting billets to neighbouring countries (Nepal, Bhutan, Bangladesh, Sri Lanka) and African markets.

7. Dealer Network Expansion

Strengthen presence in Tamil Nadu and Telangana markets through dealer onboarding, site visits, and builder engagement programs.

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2. IS 1893:2025 - Criteria for earthquake resistant design of structures (Part 1: General provisions and buildings)
3. IS 13920 - Ductile detailing of reinforced concrete structures subjected to seismic forces
4. IS 2830 / IS 2831 - Carbon steel billets, blooms, and slabs for re-rolling

Text Books & References:

1. Chandra, B. (2022). *Steel Making*. Tata McGraw-Hill Education.
2. Ghosh, A. & Chatterjee, A. (2021). *Iron Making and Steelmaking: Theory and Practice*. PHI Learning.
3. Bureau of Indian Standards (BIS). (2008). **Handbook on TMT Bars - IS 1786**. New Delhi: BIS Publication.

Websites & Online Resources:

1. Bureau of Indian Standards (BIS) - www.bis.gov.in
2. Ministry of Steel, Government of India - www.steel.gov.in
3. Joint Plant Committee (JPC) - www.jpcindiansteel.com

4. World Steel Association - www.worldsteel.org
5. Kamdhenu Group - www.kamdhenugroup.com

Internal Company Documents (provided during internship):

1. Size_and_bundle_chart.xlsx - TMT bar dimensional and bundle weight data
2. 5-day purchase, production & sales data (May 1-5, 2026)
3. Spectro Lab test reports (various dates)
4. Tensile test records (Grade Testing Laboratory)
5. Daily production logs and dispatch registers

APPENDICES

Appendix A: Photographs from Internship

(Insert the following photos from your "photos from industry.docx" file in this section, with the titles as specified)

Photo A.1: CCM Casting Billet

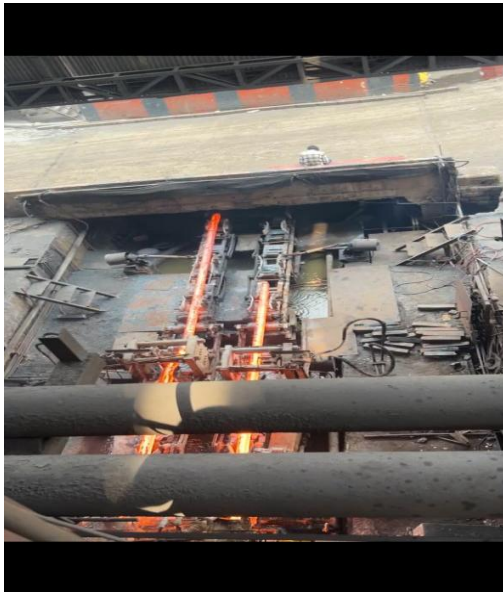


Photo A.2: TMT Rolling Mill Processing



Photo A.3: Molten Metal to CCM



Photo A.4: Me with General Manager (Mr. Pradip Kumbhakar)



Photo A.5: Molten Metal to CCM (2)



Photo A.6: My Internship Working Industry and Company (Plant View)



Photo A.7: Path Directly from CCM to Rolling Mill

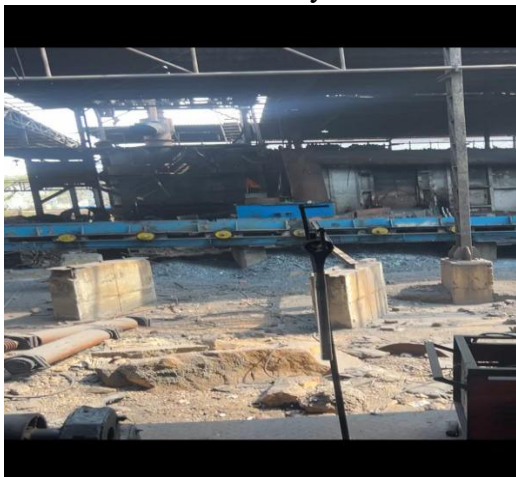


Photo A.8: Pouring of Liquid Molten Metal into CCM

Photo A.9: Scrap Yard and Scrap Collection



Photo A.10: Stockyard and TMT Bars Loading into Vehicles for Dispatch



Photo A.11: TMT Rolling Mill Processing



Photo A.12: With General Manager and HR Department Head, and Billing Department Executive



Appendix B: TMT Bar Size & Bundle Reference Chart

Size	Dia (mm)	Radius (mm)	Area (mm ²)	Volume (g/mm)	Wt/m (kg)	Wt/11m (kg)	Qty/Bundle	Bundle Wt (kg)	IS 1786 Tol.
8mm	8	4	50.24	0.3944	0.395	4.345	10	43.45	±7%
10mm	10	5	78.50	0.6162	0.617	6.787	7	47.51	±7%
12mm	12	6	113.04	0.8874	0.889	9.779	5	48.90	±5%
16mm	16	8	200.96	1.5775	1.580	17.38	3	52.14	±5%
20mm	20	10	314.00	2.4649	2.469	27.159	2	54.32	±3%
25mm	25	12.5	490.63	3.8514	3.858	42.438	1	42.44	±3%
32mm	32	16	803.84	6.3101	6.321	69.531	1	69.53	±3%

Appendix C: Key Mechanical Properties Observed (Tensile Test)

Parameter	Recorded Value	IS 1786 Requirement (Fe 500D)	Status
Yield Stress	542 N/mm ²	≥ 500 N/mm ²	PASS
Tensile Strength	644 N/mm ²	≥ 545 N/mm ²	PASS
TS/YS Ratio	1.20	≥ 1.10	PASS
Elongation %	18-10%	≥ 16% (for Fe 500D)	PASS